



Robot Simulator

Coding Task HiQ

Recruitment Process

DESCRIPTION

The application is a simulation of a toy robot moving on a square tabletop, of dimensions 5 units x 5 units. There are no other obstructions on the table surface. The robot is free to roam around the surface of the table, but must be prevented from falling to destruction. Any movement that would result in the robot falling from the table must be prevented, however further valid movement commands must still be allowed.

ASSIGNMENT

Create an application that can read commands of the following form:

PLACE,X,Y,F Put the robot on the table in position X, Y and facing NORTH, SOUTH, EAST or WEST

MOVE Move the robot one unit forward in the direction it is currently facing

LEFT Rotate the robot 90 degrees left

RIGHT Rotate the robot 90 degrees right

REPORT Announce the X, Y and F of the robot. This can be in any form, but standard output is sufficient

- The origin (0,0) can be considered to be the SOUTH WEST most corner.
- The first valid command to the robot is a PLACE command, after that, any sequence of commands may be issued, in any order, including another PLACE command. The robot should ignore all commands in the sequence(MOVE, LEFT, RIGHT and REPORT) until a valid PLACE command has been executed.
- Input should be in the form of a text file.
- Provide test data to exercise the application.
- The solution should be sufficiently unit tested.
- It is not required to provide any graphical output showing the movement of the toy robot.

CONSTRAINTS

- The toy robot must not fall off the table during movement. This also includes the initial placement of the toy robot. Any move that would cause the robot to fall must be ignored.

DELIVERABLES

- The source files, without any dependencies except for test dependencies.
- The test data and any test code.

EXAMPLES

a)

PLACE,0,0,NORTH

MOVE

REPORT

Output: 0,1,NORTH

b)

PLACE,0,0,NORTH

LEFT

REPORT

Output: 0,0,WEST

c)

PLACE,1,2,EAST

MOVE

MOVE

LEFT

MOVE

REPORT

Output: 3,3,NORTH

GOOD LUCK!